FINAL YEAR PROJECT / LOGBOOK

Autonomous Robot for Assessing Wheelchair Accessibility

Student Enze TU
Supervisor Steve Greedy

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1 Week 1

1.1 Executive summary

The project proposes a compact autonomous or semi-autonomous robot for assessing wheelchair accessibility in buildings. It will collect spatial evidence, identify potential barriers and provide a report and live web view, following the project brief [1].

WHAT? An indoor inspection system that maps selected routes, measures openings and clearances, locates obstructions and presents findings against an agreed wheelchair envelope. Each finding should include its location, measurement and uncertainty.

WHY? Narrow openings, restricted turning space and temporary obstacles can interrupt an otherwise usable route. Repeatable, location-linked evidence can support inspection and decisions about improvements.

HOW? Start with a supervised four-wheel-drive platform (TODO:Check the specific terrain), then add mapping, measurements and goal-based navigation. Terrain and threshold capability remain to be confirmed. A camera and UWB positioning are optional additions, subject to usefulness, equipment and time.

1.2 WHAT: project scope

The robot should inspect corridors, doorways and manoeuvring areas within the agreed wheelchair-related spatial constraints. Its outputs are an annotated map, a list of potential barriers and a report linked to the recorded observations. Initial measurements concern widths, clearances and obstacle positions; vertical features require additional sensing and validation.

The robot's footprint does not represent every wheelchair: passing through a gap alone does not establish accessibility. Report results as measured, uncertain or not assessed, using the agreed wheelchair dimensions and clearance margin.

Table 1. Four fundamental subsystems expressed as project functions.

Subsystem	Responsibility	Intended output
Moving vehicle	Move safely through the test environment under supervised control	Recorded inspection route and stop status
Measurement / mapping	Observe geometry and locate potential access barriers	Spatial map and measured features
Data collection	Retain observations, locations and test conditions	Replayable, traceable inspection record
Data presentation	Explain findings to an operator or reviewer	Live web view and exported report

1.2.1 Functional requirements

Table 2. Application requirements and proposed verification.

ID	Required outcome	Evidence
R1	Operate within an agreed physical and turning envelope	Dimensional and manoeuvring checks
R2	Stop safely on emergency input or expired commands	Restrained fault and stop trials
R3	Survey and locate selected indoor features	Map and reference-location checks
R4	Measure widths and identify geometric obstructions	Independent dimensional survey
R5	Follow a selected inspection route with supervision	Repeated route / goal trials
R6	Preserve observations and environmental conditions	Recorded run and replay check
R7	Present potential issues through a live web interface	Timestamped map and issue display
R8	Export a report with uncertainty and coverage limits	Traceable sample inspection report

1.2.2 Autonomy and scope

Teleoperation is used during integration. **Goal-based navigation** lets the robot plan and follow a route to operator-selected goals. **Autonomous exploration** also selects where to inspect next and is a later scope decision. The initial assessment covers selected widths, clearances and obstacle locations; stairs, transport of a person and unsupervised public operation are excluded.

1.3 WHY: value for wheelchair accessibility

The assessment should explain where access may be restricted and why: insufficient opening width, limited turning space or an obstruction along the route. Recording the location, conditions and uncertainty makes the evidence useful for later review [1].

1.3.1 Limits of the assessment

Geometry alone cannot assess door-opening effort, surface comfort or individual needs. A horizontal scan cannot reliably measure ramps, steps or overhead clearance. The indoor environment and required barrier types remain to be agreed; results are not accessibility certification. (TODO:Determine the indoor environment)

1.4 HOW: proposed implementation

1.4.1 Four-wheel platform and power

Four encoded gearmotors drive the left and right wheel pairs for skid steering. Mount the battery low, balance the payload and keep sensor mounts rigid. Confirm terrain, grip and turning behaviour; four driven wheels alone do not guarantee stability.

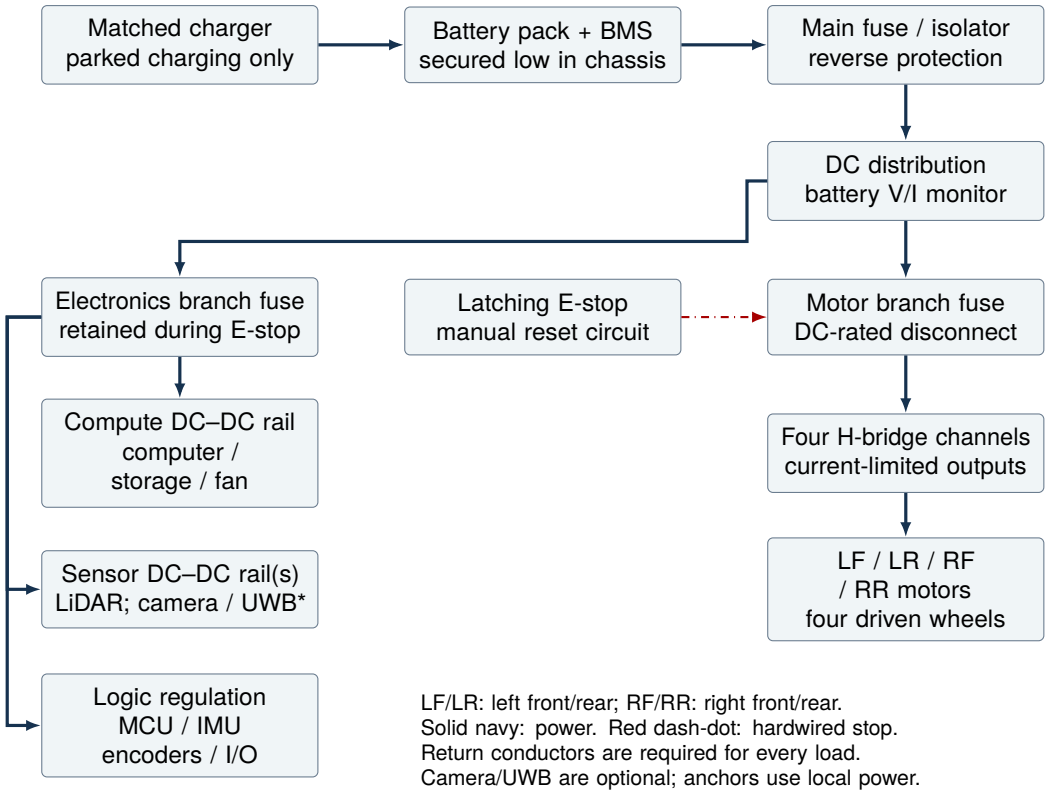


Figure 1. Proposed power architecture, not a pin-level wiring schematic. Charging is a separate parked operating state; motor operation is inhibited while charging.

Before wiring: confirm rail voltages, four-motor start/stall loads, fuse and cable ratings, return paths, thermal margin and braking transients. Test charging inhibit, emergency stopping and manual re-arm. (TODO)

1.4.2 Hardware data, control and safety interfaces

The MCU handles four wheel-speed loops and command timeouts; the computer handles mapping and navigation. Calibrate skid-steer odometry on the intended floor. Dashed-outline sensor boxes are optional.

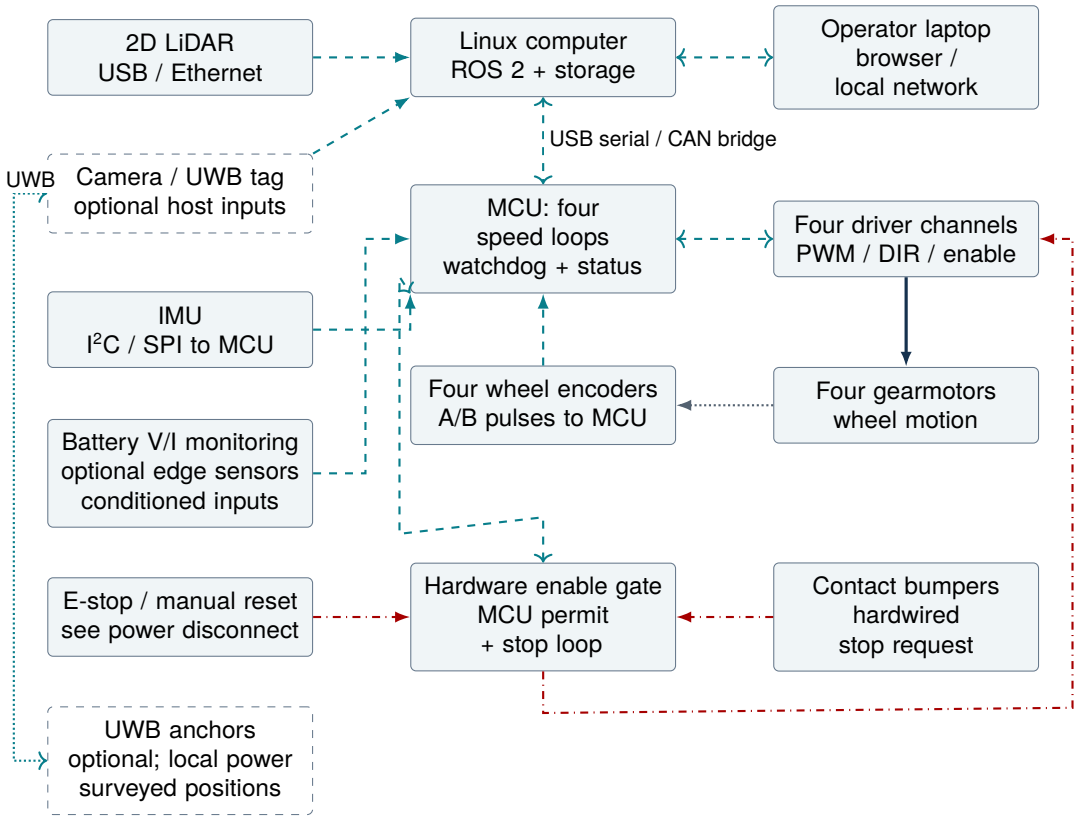


Figure 2. Data/control (teal dashed), hardwired stop (red dash-dot), motor power (navy solid) and mechanical feedback (grey dotted). Dotted radio link and dashed-outline camera/UWB boxes are optional; anchors use local power.

Interface checks: define command units/timeouts, pin levels, encoder signs, fault reporting and default-disabled motor enable. Hardware stopping must work without the computer. Restrict initial trials to protected level floors; validate edge sensing before exposed-edge trials.

1.4.3 ROS 2 software and information flow

Use ROS 2 Jazzy with compatible drivers; confirm topic names, timestamps and command types during integration [5].

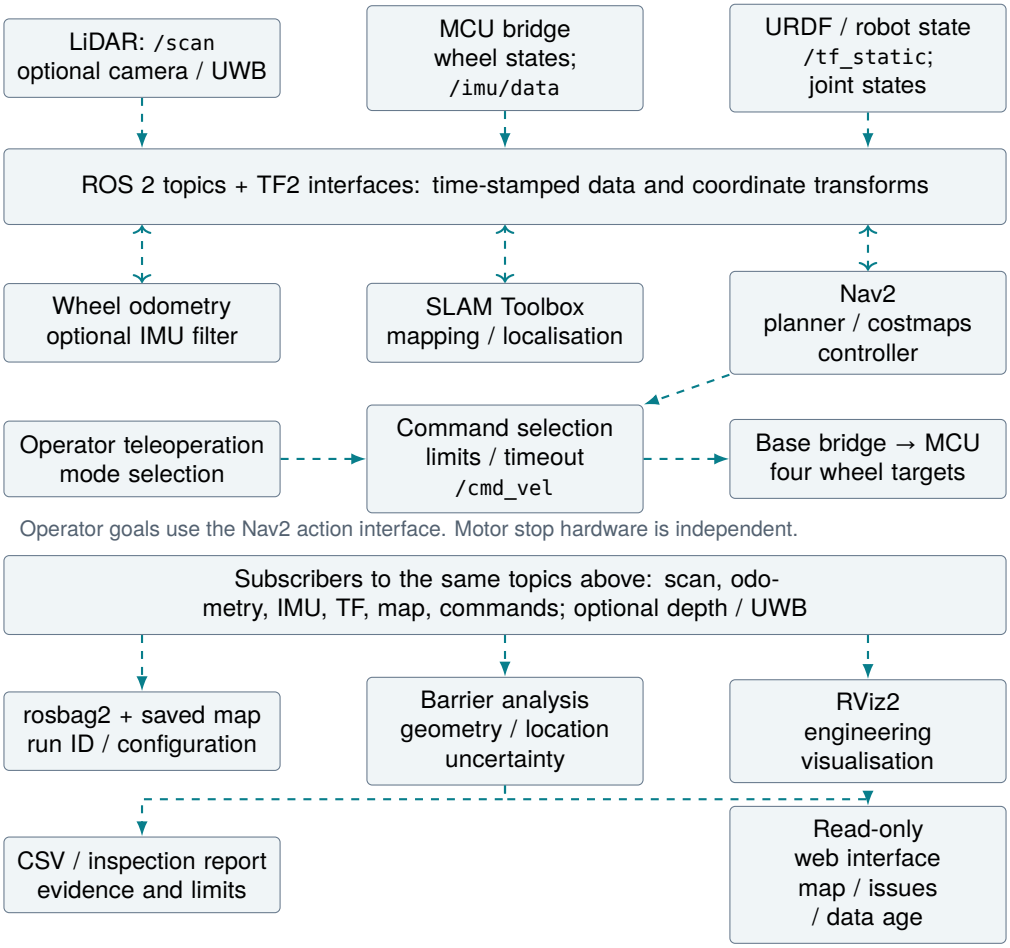


Figure 3. Mapping and navigation above; recording, assessment and presentation below. The lower panel subscribes to outputs above. Dashed teal arrows denote data/control, not power.

Node	Reads	Publishes / owns
Local estimator	Wheel feedback; optional IMU	`/odom`; `odom -> base_link`
SLAM Toolbox	`/scan`; timed TF	`/map`; `map -> odom`
Nav2	Map, scan, odometry, TF; goals	Navigation velocity request

Each dynamic TF edge has one publisher [3, 4, 6]. Record topics with `rosbag2` and retain calibration/settings separately [10]. Camera and UWB inputs are optional. Log UWB observations for evaluation first; any fusion requires frame alignment and a validated estimator, not a direct connection to SLAM Toolbox.

1.4.4 Hardware inventory and procurement

Ask: confirm laboratory stock; buy if unavailable. **Buy/check**: check stores before ordering. **Optional**: discuss before selection. No laboratory availability is confirmed. Check kit contents to avoid duplicate orders; select electrical ratings after the motor and sensor loads are known.

Table 3. Combined hardware list, grouped by function.

ID	Hardware	Qty.	Function / key check	Status
A. Platform and drive — motion and stability				
H01	Four-wheel chassis	1	Wheelchair envelope; low battery position	Ask
H02	Geared drive motors	4	Matched torque/speed; known stall current	Ask
H03	Wheel encoders	4	Speed feedback; compatible signal levels	Ask
H04	Wheels, tyres and hubs	4 sets	Load rating, grip and shaft fit	Ask
H05	Motor mounts / bearings	1 set	Secure drive assembly; check kit contents	Buy/check
H06	Motor drivers	4 ch.	Reversible drive; current limit and enable	Ask
B. Computing and sensing — control and observation				
H07	Motor-control MCU	1	Four speed loops, encoder inputs, watchdog	Ask
H08	Linux computer	1	Compatible drivers and compute capacity	Ask
H09	Storage / cooling	1 set	OS, recorded data and heat management	Buy/check
H10	2D LiDAR	1	Planar mapping; range and driver support	Ask
H11	IMU	1	Motion observations; axes, bias and timing	Ask
H12	RGB-D camera	1	Depth / visual barriers; calibration needed	Optional
H29	UWB tag + anchor kit	1 kit	Anchored position; coverage, survey and NLOS tests	Optional
H15	Sensor mounts / enclosure	1 set	Rigid alignment, ventilation and protection	Buy/check
C. Safety and power — supply, protection and stopping				
H13	Contact bumpers	1 set	Front/rear contact stop input	Buy/check
H14	Drop-off sensors	1 set	Edge detection; validate surface response	Optional
H16	Battery + BMS	1 pack	Energy, peak discharge and secure mounting	Ask
H17	Charger / connector	1 set	Match battery; inhibit motion while charging	Ask
H18	DC–DC regulators	As needed	Compute/sensor/logic rails and current margin	Buy/check
H19	Fuses / DC distribution	1 set	Branch protection, isolator and returns	Buy/check
H20	E-stop / disconnect / reset	1 set	Hardware stop, enable gate and re-arm	Buy/check
H21	Voltage / current monitor	1 set	Battery and driver status within input limits	Buy/check
D. Interfaces and test resources — integration and verification				
H22	Wiring / connectors	1 set	Rated cables, crimps, labels and strain relief	Buy/check
H23	Level / bus adapters	As needed	Signal conversion or CAN/USB interface	Buy/check
H24	Network / data cables	1 set	Robot communication and sensor data links	Ask
H25	Laptop / controller	1 set	Development and supervised teleoperation	Ask
H26	Supply / DMM / scope	Shared	Current-limited commissioning and diagnosis	Ask lab
H27	Reference survey tools	Shared	Independent distance and angle checks	Ask lab
H28	Assembly / test materials	1 set	Fasteners, fabrication, stands and barriers	Ask lab

1.4.5 Coordinate frames and sensor integration

Use SI units and the ROS body convention: x forward, y left and z up, with positive yaw counter-clockwise when viewed from above [7]. The basic frame chain is

$$\texttt{map} \longrightarrow \texttt{odom} \longrightarrow \texttt{base_link} \longrightarrow \{\texttt{laser}, \texttt{imu_link}\}. \tag{1}$$

The meanings follow REP-105 [8]: `map` is the global mapping reference and may exhibit correction jumps; `odom` is locally continuous but can drift; `base_link` is fixed to the chassis. Each sensor has a frame at its measurement origin.

Integration checks URDF defines robot geometry and fixed sensor mounts; TF2 provides transforms at observation times [9]. Check wheel dimensions and encoder signs, sensor axes, extrinsics, units and covariances. Use acquisition timestamps and compatible topic QoS; replay with a consistent clock.

If fitted, calibrate camera intrinsics/extrinsics. UWB requires surveyed anchors, a tag offset and alignment to the map frame; multipath and non-line-of-sight errors must be evaluated [2]. Neither option is required for the baseline.

1.4.6 Experimental validation plan

Agree tolerances and safety limits before testing. Start with five repeats per condition and refine the trial count from pilot variability; retain failed and interrupted runs.

Table 4. Proposed experiments and measurable outcomes.

Experiment	Method and independent reference	Metrics
Four-wheel motion	Straight, turning and closed routes on the test floor; independent start/end survey	Endpoint drift; heading error; wheel slip and repeatability
LiDAR accuracy	Stationary targets at measured ranges and several incidence angles/materials	Range bias, RMSE, spread and invalid-return fraction
Map geometry	Compare mapped wall separation and room dimensions with a tape or laser-distance survey	Dimension error; wall residuals; loop inconsistency
Navigation	Fixed goals and documented obstacle layouts; predefined timeout and goal tolerance	Success fraction; time; path length; interventions and collisions
Environmental dimensions	Corridors and door openings measured independently at corresponding heights	Width bias, absolute error, repeatability and interval coverage
Stop / stability	Restrained stop tests, loss of commands and low-speed turns with final payload	Stop distance/time; re-arm behaviour; observed wheel lift or movement
Report / live display	Known test barriers, traceable run records and deliberate data interruption	Issue-location agreement; report completeness; display age and stale-data flag

1.5 Questions

1. Which wheelchair dimensions and turning envelope should the accessibility assessment use?
2. Which terrain and barriers should be assessed: doorways, turns, obstacles, thresholds, ramps, steps, uneven surfaces or overhead obstructions? Should the robot cover a broad range or a selected set? Must it physically traverse these features, or only detect and assess them? If step climbing is required, what maximum height should guide the choice of wheels or tracks?
3. Should the robot explore unfamiliar environments without prior setup, or use a designated test area where UWB anchors can be installed? If optional UWB is included, should it supplement LiDAR SLAM or provide the primary localisation method?
4. Does the laboratory have an onboard computer, such as a Jetson Orin NX, available for SLAM, navigation and sensor processing, or should processing run on an external computer over a network?
5. What is the maximum hardware budget, and which platforms, sensors and other items in H01–H29 are available to borrow from the laboratory?
6. Is a camera needed for depth measurements or object recognition, or is planar geometry sufficient initially?
7. What information must the real-time web interface display?

References

[1] Steve Greedy. *Autonomous Robot for Assessing Accessibility* [*128*]. University of Nottingham project brief, p. 284, undated. Local source: `reference/project_description.pdf`.

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